In [4]:
```
print("Name    : Khush Pithva\nRoll No.:24BCE174" )
```

```
Name    : Khush Pithva
Roll No.:24BCE174
```

In [ ]:
```
.x=5
y="ABC"
print(x)
print(y)
x="python"
y="is"
z="a programming language"
print(x,y,z)
print(type(x))
```

```
5
ABC
python is a programming language
<class 'str'>
```

In [ ]:
```
#input from user
x=int(input("Enter a number x:"))   #we have to type cast here or it woul
y=int(input("Enter a number y:"))   #consider input as string.
print(x+y)
print(type(x))
a=eval(input("enter values:")) #give commas while giving input else error
a+=a                           #be given.
print(a)
```

```
Enter a number x:2
Enter a number y:3
5
<class 'int'>
enter values:2,3,4
(2, 3, 4, 2, 3, 4)
```

In [ ]:
```
#numbers.
#integer
s=23;
print(s)
s=-23;

#float
print(s)
s=2.3
print(s)
s=10e4+3;

#complex
print(s)
s=10e3+3.23j
print(s)
s=2.3+3j
print(s)
```

```
23
-23
2.3
100003.0
(10000+3.23j)
(2.3+3j)
```

In [5]:
```
#strings(immutable)
s=' khush '
print(s)
print(s[0])
print(s[2:4])
print(s+" pithva")
print(s[-1])
print(s[-5:])
t="shree ram"
print(s+" "+t)
print(s*3)   #can't multiply string to string.
print(" ")
string1="Hello world"
print("string with the use of single quotes : ")
print(string1)
s2=("python programming")
print("\nstring with the use of double quotes : ")
print(s2)
print(type(s2))
s2="Hello world and pytrhon programming"
print("\nstring with the use of triple quotes : ")
print(type(s2))
```

```
 khush

hu
 khush  pithva

hush
 khush  shree ram
 khush  khush  khush

string with the use of single quotes :
Hello world

string with the use of double quotes :
python programming
<class 'str'>

string with the use of triple quotes :
<class 'str'>
```

In [2]:
```
#list [] (mutable)
#general(basic).
s=["khush","pithva","bharatbhai"]
print(s[1])
s[1]='shree ram'
t=['hello','bye','tie']
print(s)
print(s[1:])
print(s+t)
print(s*2)
print(s[-1])
```

```
print(s[:-1])
print(" ")

#nested list.
s=[['khush','pithva','bharatbhai'],'shree ram','hello']
v=[['priyansh','patel'],'shree ram']
print(s[0])
print(s[0][1])
print(s+v)
print(s[0][1:])
print(s[0][1:],s[1:])    #imp
print(s[0:][1:])

#muting
print('\n')
v[1]='gandu'
print(v)
v[0][1]='gandu pro max'
print(v)

#adding list
print('\n')
t=['khush','lathu','lalo']
t.append('dhyey')
print(t)
t.insert(2,'priyansh')
print(t)
t.extend(v)
print(t)

#removing from list
print('\n')
t.remove('lalo')
print(t)
t.pop(1)
print(t)
del t[0:2]
print(t)
t.clear()
print(t)
```

```
pithva
['khush', 'shree ram', 'bharatbhai']
['shree ram', 'bharatbhai']
['khush', 'shree ram', 'bharatbhai', 'hello', 'bye', 'tie']
['khush', 'shree ram', 'bharatbhai', 'khush', 'shree ram', 'bharatbhai']
bharatbhai
['khush', 'shree ram']

['khush', 'pithva', 'bharatbhai']
pithva
[['khush', 'pithva', 'bharatbhai'], 'shree ram', 'hello', ['priyansh', 'pa
tel'], 'shree ram']
['pithva', 'bharatbhai']
['pithva', 'bharatbhai'] ['shree ram', 'hello']
['shree ram', 'hello']


[['priyansh', 'patel'], 'gandu']
[['priyansh', 'gandu pro max'], 'gandu']


['khush', 'lathu', 'lalo', 'dhyey']
['khush', 'lathu', 'priyansh', 'lalo', 'dhyey']
['khush', 'lathu', 'priyansh', 'lalo', 'dhyey', ['priyansh', 'gandu pro ma
x'], 'gandu']


['khush', 'lathu', 'priyansh', 'dhyey', ['priyansh', 'gandu pro max'], 'ga
ndu']
['khush', 'priyansh', 'dhyey', ['priyansh', 'gandu pro max'], 'gandu']
['dhyey', ['priyansh', 'gandu pro max'], 'gandu']
[]
```

In [3]:

```
#tuple () (immutable)
s=(1,23,4,2,5,6)
print(s[0])
print(s[2:])
t=('khush',34)
print(s+t)
print(s[-5:-2])
print(s[:-1])
print(s[:])
print(s)
print(" ")
t1=('a','b','c','d')
t2=('e','f','g','h')
t3=('i','j')          #('b',) = tuple    otherwise   ('b') = string
l=['hi','hello','bye']
print(tuple(l))
print(t1)
print(t2+t3)
print(t3[1])
print(t2[1:])

#muting by converting
print(" ")
t1=('a','b')
y=list(t1)
y[1]='c'
t1=tuple(y)
```

```
print(t1)

#add
print(" ")
t1=('a','b')
y=list(t1)
y.append('c')
t1=tuple(y)
print(t1)
```

```
1
(4, 2, 5, 6)
(1, 23, 4, 2, 5, 6, 'khush', 34)
(23, 4, 2)
(1, 23, 4, 2, 5)
(1, 23, 4, 2, 5, 6)
(1, 23, 4, 2, 5, 6)

('hi', 'hello', 'bye')
('a', 'b', 'c', 'd')
('e', 'f', 'g', 'h', 'i', 'j')
j
('f', 'g', 'h')

('a', 'c')

('a', 'b', 'c')
```

In [ ]:

```
#dictionary(mutable)
d1={1:'a',2:'b',3:'c',4:'d'}
print(d1)
print(d1.keys())
print(d1.values())

#key values
print(" ")
print(d1[1])
print(d1.get(3))

#add
print(" ")
d2={5:'e',6:'f',7:'g'}
d1.update(d2)
print(d1)

#delete
print(" ")
del d1[2]
print(d1)

#clear
print(" ")
d1.clear()
print(d1)
```

```
{1: 'a', 2: 'b', 3: 'c', 4: 'd'}
dict_keys([1, 2, 3, 4])
dict_values(['a', 'b', 'c', 'd'])

a
c

{1: 'a', 2: 'b', 3: 'c', 4: 'd', 5: 'e', 6: 'f', 7: 'g'}

{1: 'a', 3: 'c', 4: 'd', 5: 'e', 6: 'f', 7: 'g'}

{}
```

In [ ]:

```
#operators
a=6
b=5

#Arithmetic operators
print(a+b)
print(a-b)
print(a*b)
print(a/b)
print(a%b)
print(a**b)  #power
print(a//b)  #greatest integer.

#Comparison operators
print(" ")
print(a==b)
print(a!=b)
print(a>b)
print(a<b)
print(a<=b)
print(a>=b)

#assigment operators
print(" ")
c=a+b
print(c)
c+=a
print(c)
c-=a
print(c)
c*=a
print(c)
c/=a
print(c)
c**=a
print(c)
c//=a
print(c)

#Bitwise Operators
print(" ")
print(a&b)                          #and
print(a|b)                          #or
print(a^b)                          #xor
print(~a)                           #not
print(a<<2)                         #left shift
print(a>>2)                         #right shift
```

```
#Logical Operators
print(" ")
print(a and b)
print(a or b)
print(not(a and b))
a=True
b=False
print(a and b)
print(a or b)
print(not(a and b))

#Membership Operators
print(" ")
L1=['a','b','c','d','e']                    # use in string , list and tup
print('a' in L1)
print(3 in L1)
print(3 not in L1)
print('a' not in L1)

#Identity Operators
print(" ")
a=10
b=10
print(a is b)
print(a is not b)
a='c'
b=c
print(a is not b)
```

```
11
1
30
1.2
1
7776
1

False
True
True
False
False
True

11
17
11
66
11.0
1771561.0
295260.0
```

In [ ]:

```
#if else
value=1
if value==0:
  print("0")
else:
  print(value)
```

```
value=10;
if value>5:
  if value>10:
    print("greater than 10")
  else:
    print("greater than 5 but less than 10")
else:
  print("less than 5")

a=int(input())
b=int(input())
if a<b:
  print("b is greater than a")
elif b<a:
  print("a is greater than b")
else:
  print("a is equal to b")
```

```
1
greater than 5 but less than 10
6
5
a is greater than b
```

In [ ]:

```
#while loops
a=int(input())
while a!=0:
  print("T_V_")
  a=a-1;
print("good bye");
```

```
8
T_V_
T_V_
T_V_
T_V_
T_V_
T_V_
T_V_
T_V_
good bye
```

In [6]:

```
#for loop
a=[input()]
for given in a:
  print(given)
print(" ")
print("loop over")

l=['khush','priyansh','dhyey','fadu','priyansh']
for full in l:
  print(full)

print(" ")

for index in range(len(l)):
  print(index)
  print(l[index])

t=0
n=int(input("How many numbers will you enter? "))
```

```
for _ in range(n):
  num=int(input("Enter a number : "))
  t +=num

print("The total number is : ",t)
```

```
khush

loop over
khush
priyansh
dhyey
fadu
priyansh

0
khush
1
priyansh
2
dhyey
3
fadu
4
priyansh
The total number is :  8
```

In [4]:

```
passing_Score = 33
my_Score = int(input("My score is:"))
if(my_Score >= passing_Score):
    print("Congratulations! You passed in the exam")
else:
    print("Sorry! You failed the exam, better luck next time")
```

```
Congratulations! You passed in the exam
```

In [3]:

```
student_name = input("Enter the name:")
marks = {'Jainil': 380, 'Drashti': 382, 'Zeel': 382, 'Prachi':373 ,'Priya
for student in marks:
    if student == student_name:
        print(marks[student])
        break
else:
    print('No entry with that name found.')
```

```
No entry with that name found.
```

In [7]:

```
for i in range(1,6):
    print("\n")
    for j in range(1,i+1):
        print(j,end=" ")
```

```
1

1 2

1 2 3

1 2 3 4

1 2 3 4 5
```

In [6]:
```
for i in range(5, 0, -1):
    for j in range(1, i + 1):
        print(j, end=" ")
    print()
```

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

In [5]:
```
for i in range(1,6):
    print("\n")
    for j in range(1,i+1):
        print("*",end=" ")
```

```
*

* *

* * *

* * * *

* * * * *
```

In [8]:
```
dob=str(input("enter your dob : "))
day=int(dob[0:2])
month=int(dob[3:5])
year=int(dob[6:])
if month>2:
    year=year+1
    month = (month+10) - 12
else:
    month=month+10

print("Day:", day)
print("Month:", month)
print("Year:", year)
```

```
Day: 1
Month: 1
Year: 24
```

In [7]:
```
#break & continue
for name in "priyansh dhyey baba":
  if name=='e':
     break
  print(name)
```

```
print(" ")
for name in "priyansh dhyey baba ki":
  if name=='y':
    continue
  print(name)
```

```
p
r
i
y
a
n
s
h

d
h
y

p
r
i
a
n
s
h

d
h
e

b
a
b
a

k
i
```

In [2]:
```
#strings revisited.
import math
x=2.34
print(int(x))
print(float(x))
y=3
print(complex(x,y))
print(int(abs(x)))
print(math.ceil(x))   #math.ceil kam nai aaptu.
```

```
2
2.34
(2.34+3j)
2
3
```

In [ ]:
```
#Function
def tv():
  print("priyansh kaka")
tv()

def tv(n1,n2):
```

```
  s=n1+n2
  print(s)
tv(int(input()),int(input()))
```

```
jay kaka
2
3
5
```

In [1]:
```
import numpy as np
a=np.array([1,2,3,4,5])
print(a)
```

```
[1 2 3 4 5]
```

In [2]:
```
from numpy import*
a=array([[1,2,3,4,5],[6,7,8,9,10]]) #2D array
print(a)
```

```
[[ 1  2  3  4  5]
 [ 6  7  8  9 10]]
```

In [ ]:
```
#numpy
from numpy import *
a=array([[1,2],[3,4]])
print(a.shape)
print(a.ndim)
print(a.dtype.name)
print(a.itemsize)
print(a.size)
print(type(a))
print(ndarray)

print(" ")

a=array([[2,3],[4,5],[6,7]])
print(a.shape)
print(a.ndim)
print(a.dtype.name)
print(a.itemsize)
print(a.size)
print(type(a))
print(ndarray)

a=array([10,20,30,'khush'])  #string ma badha convert thase.
print(a)
a=array([10,20,30,40.5])   #badha decimal ma convert thase
print(a)
```

```
(2, 2)
2
int64
8
4
<class 'numpy.ndarray'>
<class 'numpy.ndarray'>

(3, 2)
2
int64
8
6
<class 'numpy.ndarray'>
<class 'numpy.ndarray'>
['10' '20' '30' 'tanuj']
[10.  20.  30.  40.5]
```

In [ ]:
```
#Reshape
from numpy import*
d=array([[2,3,4,5],[6,7,8,9],[10,11,12,13]])
print(d)
d=d.reshape(4,3)
print(d)

#empty
print('\n')
x=empty([2,4],dtype=int)
y=empty([3,4],dtype=float)
print(x)
print(y)
```

```
[[ 2  3  4  5]
 [ 6  7  8  9]
 [10 11 12 13]]
[[ 2  3  4]
 [ 5  6  7]
 [ 8  9 10]
 [11 12 13]]


[[50201489        0        0        0]
 [       0        0        0        0]]
[[4.21969319e-316 0.00000000e+000 0.00000000e+000 4.00193173e-322]
 [4.17472506e-316 6.59872039e-310 7.85138443e-313 4.10074486e-322]
 [0.00000000e+000 0.00000000e+000 1.36868729e-311 7.85138443e-313]]
```

In [ ]:
```
# Mathematical operator on arrays
from numpy import*
a=array([10,18,15,22])
a=a+3
print(a)
a=a-3
print(a)
a=a*3
print(a)
a=a/3
print(a)
```

```
[13 21 18 25]
[10 18 15 22]
[30 54 45 66]
[10. 18. 15. 22.]
```

In [ ]:
```
#Linespace
from numpy import*
a=linspace(0,100,3)
print(a)

#Arange
print('\n')
a=arange(5)
b=arange(2,5)
c=arange(10,1,-2)
print(a)
print(b)
print(c)

#Zeros and Ones
print('\n')
j=zeros(3)
k=ones(3)
print(j)
print(k)

#eye
print('\n')
a=eye(3)
print(a)
```

```
[  0.  50. 100.]


[0 1 2 3 4]
[2 3 4]
[10  8  6  4  2]


[0. 0. 0.]
[1. 1. 1.]


[[1. 0. 0.]
 [0. 1. 0.]
 [0. 0. 1.]]
```

In [ ]:
```
#Concatenate
from numpy import*
a=array([2,3,4,5])
b=array([6,7,8,9])
c=concatenate([a,b])
print(c)


d=array([[1,2,3],[4,5,6],[7,8,9]])
e=concatenate([d,d],axis=1)
print(e)
f=concatenate([d,d],axis=0)
print(f)
```

```
#hstack and vstack
print('\n')
a=array([[1,2,3],[4,5,6]])
b=array([[7,8,9],[10,11,12]])
c=hstack((a,b))
d=vstack((a,b))
print(c)
print(d)
```

```
[2 3 4 5 6 7 8 9]
[[1 2 3 1 2 3]
 [4 5 6 4 5 6]
 [7 8 9 7 8 9]]
[[1 2 3]
 [4 5 6]
 [7 8 9]
 [1 2 3]
 [4 5 6]
 [7 8 9]]


[[ 1  2  3  7  8  9]
 [ 4  5  6 10 11 12]]
[[ 1  2  3]
 [ 4  5  6]
 [ 7  8  9]
 [10 11 12]]
```

In [ ]:

```
from numpy import*
a=array([1,23,4,5,34,4,3,3,])
b=array_split(a,[3,6])
print(b)

c=array([[1,2,3,4,4],[1,2,34,45,64]])
d=array_split(c,3)
print(d)
print(c[1,1:4])
```

```
[array([ 1, 23,  4]), array([ 5, 34,  4]), array([3, 3])]
[array([[1, 2, 3, 4, 4]]), array([[ 1,  2, 34, 45, 64]]), array([], shape=
(0, 5), dtype=int64)]
[ 2 34 45]
```

In [ ]:

```
from numpy import*
a=array([[1,2,3,4,5,6],[7,8,9,10,11,12]])
b=array_split(a ,3 , axis=1)                          # h_split
print(b)

c=array_split(a , 3 , axis=0)
print(c)                                              # v_split

d=hsplit(a,2)
print(d)

e=vsplit(a,2)
print(e)
```

```
[array([[1, 2],
       [7, 8]]), array([[ 3,  4],
       [ 9, 10]]), array([[ 5,  6],
       [11, 12]])]
[array([[1, 2, 3, 4, 5, 6]]), array([[ 7,  8,  9, 10, 11, 12]]), array([],
shape=(0, 6), dtype=int64)]
[array([[1, 2, 3],
       [7, 8, 9]]), array([[ 4,  5,  6],
       [10, 11, 12]])]
[array([[1, 2, 3, 4, 5, 6]]), array([[ 7,  8,  9, 10, 11, 12]])]
```

In [ ]:

```python
#pandas
from pandas import *
from numpy import *
a=array([12,34,45,56,78])
s=Series(a,index=['khush','priyansh','dhyey','priyansh','fadu'])
print(s)
d=Series(7)
print(d)
e=Series(7,index=['a','b','c','d','e','f','g'])
print(e)
```

```
tanuj       12
jay         34
dhyey       45
priyansh    56
fadu        78
dtype: int64
0    7
dtype: int64
a    7
b    7
c    7
d    7
e    7
f    7
g    7
dtype: int64
```

In [ ]:

```python
#Mathematical Operations in Series
from pandas import*
s=Series([17,37,57,49,40])
a=Series([76,46,39,83,25])
b=Series([25,34,65,45])
print(s*2)
print(s**2)
print(s[s>=57])
print(s+a)                              # same index
print(a+b)
print(a.add(b,fill_value=0))
```

```
0     34
1     74
2    114
3     98
4     80
dtype: int64
0     289
1    1369
2    3249
3    2401
4    1600
dtype: int64
2    57
dtype: int64
0     93
1     83
2     96
3    132
4     65
dtype: int64
0    101.0
1     80.0
2    104.0
3    128.0
4      NaN
dtype: float64
0    101.0
1     80.0
2    104.0
3    128.0
4     25.0
dtype: float64
```

In [ ]:

```
#Head and Tail
from pandas import*
s=Series([23,45,67,43,56,46,98,25,36,47,56])
print(s.head())                                    #By default 5
print(s.head(3))
print(s.head(8))
print(s.tail())
print(s.tail(3))
```

```
0    23
1    45
2    67
3    43
4    56
dtype: int64
0    23
1    45
2    67
dtype: int64
0    23
1    45
2    67
3    43
4    56
5    46
6    98
7    25
dtype: int64
6     98
7     25
8     36
9     47
10    56
dtype: int64
8     36
9     47
10    56
dtype: int64
```

In [ ]:

```
# loc and iloc
from pandas import*
s=Series([23,56,84,36,57,47,36,38,98])
print(s.loc[:4])                                        # consider as labe
print(s.loc[2:8])
print(s.iloc[:4])                                       # consider as inde
print(s.iloc[4:])
```

```
0    23
1    56
2    84
3    36
4    57
dtype: int64
2    84
3    36
4    57
5    47
6    36
7    38
8    98
dtype: int64
0    23
1    56
2    84
3    36
dtype: int64
4    57
5    47
6    36
7    38
8    98
dtype: int64
```

In [ ]:

```
# loc and iloc
from pandas import*
s=Series([23,56,84,36,57,47,36,38,98])
print(s.loc[:4])                                        # consider as labe
print(s.loc[2:8])
print(s.iloc[:4])                                       # consider as inde
print(s.iloc[4:])
```

```
0    23
1    56
2    84
3    36
4    57
dtype: int64
2    84
3    36
4    57
5    47
6    36
7    38
8    98
dtype: int64
0    23
1    56
2    84
3    36
dtype: int64
4    57
5    47
6    36
7    38
8    98
dtype: int64
```

In [ ]:
```
# slicing
from pandas import*
s=Series([2,3,4,5,6,7])
print(s[3])
print(" ")
print(s[1:4])
print(s[1:4:2])
```

```
5

1    3
2    4
3    5
dtype: int64
1    3
3    5
dtype: int64
```

In [ ]:
```
# Data frame
from pandas import*
s=Series([1,2,3,4,5,6])
df=DataFrame(s)
print(df)
```

```
   0
0  1
1  2
2  3
3  4
4  5
5  6
```

In [ ]:
```
from pandas import*
name=Series(['priyansh','khush'])
branch=Series(['Cse','Cse'])
d={'Name':name,'Branch':branch}
df=DataFrame(d)
print(df)

print(" ")

l=[{'name':'priyansh','surname':'patel'},{'name':'khush','surname':'pithv
b=DataFrame(l)
print(b)
```

```
    Name Branch
0    jay    Cse
1  tanuj    Cse

       name   surname
0       jay   lathiya
1     tanuj  kanetiya
2  priyansh     patel
```

In [ ]:
```
#iterrows and itercols(aani mate khali item lakhvu)
l=[{'name':'priyansh','surname':'patel'},{'name':'khush','surname':'pithv
df=DataFrame(l)
print(df)
for(row_index,row_value) in df.iterrows():
  print('\nRow index is : ',row_index)
  print('Row value is : ',row_value)
```

```
for(cols_index,cols_value) in df.items():
  print('\nColumn index is : ',cols_index)
  print('Column value is : ',cols_value)
```

```
       name   surname
0       jay   lathiya
1     tanuj  kanetiya
2  priyansh     patel

Row index is :  0
Row value is :  name           jay
surname    lathiya
Name: 0, dtype: object

Row index is :  1
Row value is :  name          tanuj
surname    kanetiya
Name: 1, dtype: object

Row index is :  2
Row value is :  name       priyansh
surname       patel
Name: 2, dtype: object

Column index is :  name
Column value is :  0         jay
1       tanuj
2    priyansh
Name: name, dtype: object

Column index is :  surname
Column value is :  0     lathiya
1    kanetiya
2       patel
Name: surname, dtype: object
```

In [ ]:

```
#Select column print
from pandas import*
d={'Rollno.':[1,2,3,4,5,6],
   'name':['priyansh','khush','Priyansh','Krish','Dhyey','Yagndeep'],
   'surname':['patel','pithva','shree ram','mojidra','bhalani','pipaliya'
df=DataFrame(d)
df.name

df[['Rollno.','name']]
```

Out[ ]:

	Rollno.	name
0	1	Jay
1	2	Tanuj
2	3	Priyansh
3	4	Krish
4	5	Dhyey
5	6	Yagndeep

In [ ]:
```
#Rename column name
from pandas import*
s=Series([1,2,3,4])
df=DataFrame(s)
print(df)
df.columns=['number']
print(df)
df['fix number']=5
print(df)
df['sum']=df['number']+df['fix number']
print(df)
```

```
   0
0  1
1  2
2  3
3  4
   number
0       1
1       2
2       3
3       4
   number  fix number
0       1           5
1       2           5
2       3           5
3       4           5
   number  fix number  sum
0       1           5    6
1       2           5    7
2       3           5    8
3       4           5    9
```

In [ ]:
```
#Delete column
from pandas import*
d=[[1,2,3,4],[5,6,7,8],[9,10,11,12]]
df=DataFrame(d)
df.columns=['a','b','c','d']
print(df)

print('\n')
del df['d']
print(df)

print('\n')
df1=df.drop('c',axis=1)
print(df1)

print('\n')
df2=df1.drop(index=[1,2],axis=0)
print(df2)

print('\n')
print(df)
print('\n')
df3=df.pop('b')                                #error
print(df3)
```

```
   a   b   c   d
0  1   2   3   4
1  5   6   7   8
2  9  10  11  12


   a   b   c
0  1   2   3
1  5   6   7
2  9  10  11


   a   b
0  1   2
1  5   6
2  9  10


   a  b
0  1  2


   a   b   c
0  1   2   3
1  5   6   7
2  9  10  11


0     2
1     6
2    10
Name: b, dtype: int64
```

In [ ]:

```
#loc and iloc
from pandas import*
d=[[1,2,3,4],[5,6,7,8],[9,10,11,12]]
df=DataFrame(d)
df.columns=['a','b','c','d']
print(df.loc[1:2,'b'])
print('\n')
print(df.loc[1:2,'b':'c'])

print('\n')
print(df.iloc[1:2,2:3])
print('\n')
print(df.iloc[1:2,1:2])
```

```
1     6
2    10
Name: b, dtype: int64


    b   c
1   6   7
2  10  11


   c
1  7


   b
1  6
```

In [8]:
```
#head and tail
from pandas import*
d={'Rollno.':[1,2,3,4,5,6],
   'name':['priyansh','khush','Priyansh','Krish','Dhyey','Yagndeep'],
   'surname':['patel','pithva','shree ram','mojidra','bhalani','pipaliya'
df=DataFrame(d)
print(df.head())
print(" ")
print(df.tail(3))
print(" ")
print(df[0:3])
```

```
   Rollno.      name    surname
0        1  priyansh      patel
1        2     khush     pithva
2        3  Priyansh  shree ram
3        4     Krish    mojidra
4        5     Dhyey    bhalani

   Rollno.      name   surname
3        4     Krish   mojidra
4        5     Dhyey   bhalani
5        6  Yagndeep  pipaliya

   Rollno.      name    surname
0        1  priyansh      patel
1        2     khush     pithva
2        3  Priyansh  shree ram
```

In [9]:
```
#Boolen indexing
from pandas import*
d={'name':['priyansh','khush','priyansh'],
   'surname':['patel','pithva','shree ram']}
df=DataFrame(d,index=[False,True,False])
print(df)
print('\n')
print(df.loc[False])
print('\n')
print(df.iloc[1])
```

```
           name    surname
False  priyansh      patel
True      khush     pithva
False  priyansh  shree ram


           name    surname
False  priyansh      patel
False  priyansh  shree ram


name        khush
surname    pithva
Name: True, dtype: object
```

In [ ]:

```
#concat
from pandas import*
d1={'id':['1','2','3','4','5'],'value1':['A','G','F','Y','W'],'value2':['
d2={'id':['2','3','6','7','8'],'value1':['S','J','M','N','G'],'value2':['
df1=DataFrame(d1)
df2=DataFrame(d2)
df3=concat([df1,df2])
print(df3)

print('\n')
df=merge(df1,df2,on='id')
print(df)

print('\n')
df=concat([df1,df2],ignore_index=True)
print(df)

print('\n')
a={'Data1':df1,'Data2':df2}                          #To provide label
df=concat(a)
print(a)
```

```
  id value1 value2
0  1      A      T
1  2      G      H
2  3      F      V
3  4      Y      D
4  5      W      J
0  2      S      Y
1  3      J      I
2  6      M      O
3  7      N      P
4  8      G      Q


  id value1_x value2_x value1_y value2_y
0  2        G        H        S        Y
1  3        F        V        J        I


  id value1 value2
0  1      A      T
1  2      G      H
2  3      F      V
3  4      Y      D
4  5      W      J
5  2      S      Y
6  3      J      I
7  6      M      O
8  7      N      P
9  8      G      Q


{'Data1':   id value1 value2
0  1      A      T
1  2      G      H
2  3      F      V
3  4      Y      D
4  5      W      J, 'Data2':   id value1 value2
0  2      S      Y
1  3      J      I
2  6      M      O
3  7      N      P
4  8      G      Q}
```

In [ ]:

```
#Column and Row wise concat
from pandas import*
d1={'id':['1','2','3','4','5'],'value1':['A','G','F','Y','W'],'value2':['
d2={'id':['2','3','6','7','8'],'value1':['S','J','M','N','G'],'value2':['
df1=DataFrame(d1)
df2=DataFrame(d2)
df=concat([df1,df2],axis=0,ignore_index=True)
print(df)

print('\n')
df=concat([df1,df2],axis=1)
print(df)
```

```
  id value1 value2
0  1      A      T
1  2      G      H
2  3      F      V
3  4      Y      D
4  5      W      J
5  2      S      Y
6  3      J      I
7  6      M      O
8  7      N      P
9  8      G      Q


  id value1 value2 id value1 value2
0  1      A      T  2      S      Y
1  2      G      H  3      J      I
2  3      F      V  6      M      O
3  4      Y      D  7      N      P
4  5      W      J  8      G      Q
```

In [ ]:

```
#Merge operation in dataframe
from pandas import*
d1={'id':['1','2','3','4','5'],'value1':['A','G','F','Y','W'],'value2':['
d2={'id':['2','3','6','7','8'],'value1':['S','J','M','N','G'],'value2':['
d3={'id':['1','2','3','4','5','6','7','8','9','10','11'],'value':[11,12,1
df1=DataFrame(d1)
df2=DataFrame(d2)
df3=concat([df1,df2])
df4=DataFrame(d3)
df5=merge(df3,df4)
print(df5)
```

```
  id value1 value2  value
0  1      A      T     11
1  2      G      H     12
2  3      F      V     13
3  4      Y      D     14
4  5      W      J     15
5  2      S      Y     12
6  3      J      I     13
7  6      M      O     16
8  7      N      P     17
9  8      G      Q     18
```

In [ ]:

```
#Outer join
from pandas import*
d1={'id':['1','2','3','4','5'],'value1':['A','G','F','Y','W'],'value2':['
d2={'id':['2','3','6','7','8'],'value1':['S','J','M','N','G'],'value2':['
df1=DataFrame(d1)
df2=DataFrame(d2)
df=merge(df1,df2,on='id',how='outer')
print(df)

print('\n')
d1={'id':['1','2','3','4','5'],'value1':['A','G','F','Y','W'],'value2':['
d2={'id':['2','3','6','7','8'],'value1':['S','J','M','N','G'],'value2':['
df1=DataFrame(d1)
df2=DataFrame(d2)
df=merge(df1,df2,on='id',how='outer',suffixes=('_left','_right'))
print(df)
```

```
  id value1_x value2_x value1_y value2_y
0  1        A        T      NaN      NaN
1  2        G        H        S        Y
2  3        F        V        J        I
3  4        Y        D      NaN      NaN
4  5        W        J      NaN      NaN
5  6      NaN      NaN        M        O
6  7      NaN      NaN        N        P
7  8      NaN      NaN        G        Q


  id value1_left value2_left value1_right value2_right
0  1           A           T          NaN          NaN
1  2           G           H            S            Y
2  3           F           V            J            I
3  4           Y           D          NaN          NaN
4  5           W           J          NaN          NaN
5  6         NaN         NaN            M            O
6  7         NaN         NaN            N            P
7  8         NaN         NaN            G            Q
```

In [ ]:

```
#inner join
from pandas import*
d1={'id':['1','2','3','4','5'],'value1':['A','G','F','Y','W'],'value2':['
d2={'id':['2','3','6','7','8'],'value1':['S','J','M','N','G'],'value2':['
df1=DataFrame(d1)
df2=DataFrame(d2)
df=merge(df1,df2,on='id',how='inner')
print(df)

print('\n')
d1={'id':['1','2','3','4','5'],'value1':['A','G','F','Y','W'],'value2':['
d2={'id':['2','3','6','7','8'],'value1':['S','J','M','N','G'],'value2':['
df1=DataFrame(d1)
df2=DataFrame(d2)
df=merge(df1,df2,on='id',how='inner',suffixes=('_left','_right'))
print(df)
```

```
  id value1_x value2_x value1_y value2_y
0  2        G        H        S        Y
1  3        F        V        J        I


  id value1_left value2_left value1_right value2_right
0  2           G           H            S            Y
1  3           F           V            J            I
```

In [ ]:

```
#Right and left join
from pandas import*
d1={'id':['1','2','3','4','5'],'value1':['A','G','F','Y','W'],'value2':['
d2={'id':['2','3','6','7','8'],'value1':['S','J','M','N','G'],'value2':['
df1=DataFrame(d1)
df2=DataFrame(d2)
df=merge(df1,df2,on='id',how='right')
print(df)

print('\n')
d1={'id':['1','2','3','4','5'],'value1':['A','G','F','Y','W'],'value2':['
d2={'id':['2','3','6','7','8'],'value1':['S','J','M','N','G'],'value2':['
df1=DataFrame(d1)
```

```
df2=DataFrame(d2)
df=merge(df1,df2,on='id',how='left')
print(df)
```

```
  id value1_x value2_x value1_y value2_y
0  2        G        H        S        Y
1  3        F        V        J        I
2  6      NaN      NaN        M        O
3  7      NaN      NaN        N        P
4  8      NaN      NaN        G        Q


  id value1_x value2_x value1_y value2_y
0  1        A        T      NaN      NaN
1  2        G        H        S        Y
2  3        F        V        J        I
3  4        Y        D      NaN      NaN
4  5        W        J      NaN      NaN
```

In [14]:

```
import pandas as pd
runs={'TCS':{'qtr1':2500,'qtr2':2000,'qtr3':3000,'qtr4':2000},
      'WIPRO':{'qtr1':2500,'qtr2':2000,'qtr3':3000,'qtr4':2000},
      'L&T':{'qtr1':2500,'qtr2':2000,'qtr3':3000,'qtr4':2000}}

df=pd.DataFrame(runs)
print(df)
print(df.loc[:,'TCS'])
print(df.loc[:,'TCS':"WIPRO"])
```

```
      TCS  WIPRO   L&T
qtr1  2500   2500  2500
qtr2  2000   2000  2000
qtr3  3000   3000  3000
qtr4  2000   2000  2000
qtr1    2500
qtr2    2000
qtr3    3000
qtr4    2000
Name: TCS, dtype: int64
      TCS  WIPRO
qtr1  2500   2500
qtr2  2000   2000
qtr3  3000   3000
qtr4  2000   2000
```

In [15]:

```
import pandas as pd
empdata={'empid':[101,102,103,104,105,106],
         'ename':['khush','khushi','jiya','zankhana','vidhi','prinshi'],
         'doj':['12-01-2022','1-01-2022','2-08-2012','12-08-2002','2-05-2
        }
df=pd.DataFrame(empdata)
print(df)
print(df.loc[0])
df.loc[0:2]
```

```
   empid     ename         doj
0    101     khush  12-01-2022
1    102    khushi   1-01-2022
2    103      jiya   2-08-2012
3    104  zankhana  12-08-2002
4    105     vidhi   2-05-2003
5    106   prinshi  12-01-2022
empid           101
ename         khush
doj      12-01-2022
Name: 0, dtype: object
```

Out[15]:

	empid	ename	doj
0	101	khush	12-01-2022
1	102	khushi	1-01-2022
2	103	jiya	2-08-2012

In [ ]:

```
#join by index
from pandas import*
d1={'id':['1','2','3','4','5'],'value1':['A','G','F','Y','W'],'value2':['
d2={'id':['2','3','6','7','8'],'value1':['S','J','M','N','G'],'value2':['
df1=DataFrame(d1)
df2=DataFrame(d2)
df=merge(df1,df2,right_index=True,left_index=True)
print(df)
```

```
  id_x value1_x value2_x id_y value1_y value2_y
0    1        A        T    2        S        Y
1    2        G        H    3        J        I
2    3        F        V    6        M        O
3    4        Y        D    7        N        P
4    5        W        J    8        G        Q
```

In [12]:

```
import pandas as pd
empdata={'empid':[101,102,103,104,105,106],
         'ename':['khush','khushi','jiya','zankhana','vidhi','prinshi'],
         'doj':['12-01-2022','1-01-2022','2-08-2012','12-08-2002','2-05-2
        }
df=pd.DataFrame(empdata)
print(df)
print(df.head())
print(df.tail())
```

```
   empid     ename         doj
0    101     khush  12-01-2022
1    102    khushi   1-01-2022
2    103      jiya   2-08-2012
3    104  zankhana  12-08-2002
4    105     vidhi   2-05-2003
5    106   prinshi  12-01-2022
   empid     ename         doj
0    101     khush  12-01-2022
1    102    khushi   1-01-2022
2    103      jiya   2-08-2012
3    104  zankhana  12-08-2002
4    105     vidhi   2-05-2003
   empid     ename         doj
1    102    khushi   1-01-2022
2    103      jiya   2-08-2012
3    104  zankhana  12-08-2002
4    105     vidhi   2-05-2003
5    106   prinshi  12-01-2022
```

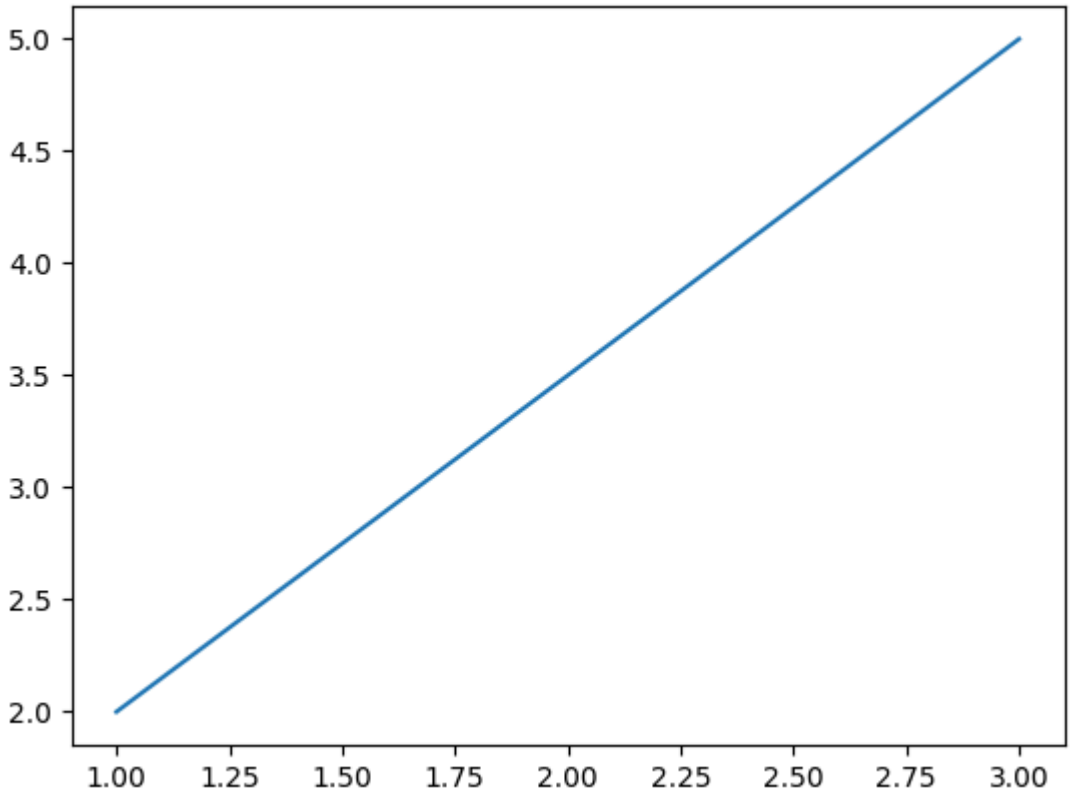
In [13]:

```
import pandas as pd
empdata={'empid':[101,102,103,104,105,106],
         'ename':['khush','khushi','jiya','zankhana','vidhi','prinshi'],
         'doj':['12-01-2022','1-01-2022','2-08-2012','12-08-2002','2-05-2
        }
df=pd.DataFrame(empdata)
print(df)
print(df.head(2))
print(df.tail(2))
print(df[2:5])
```

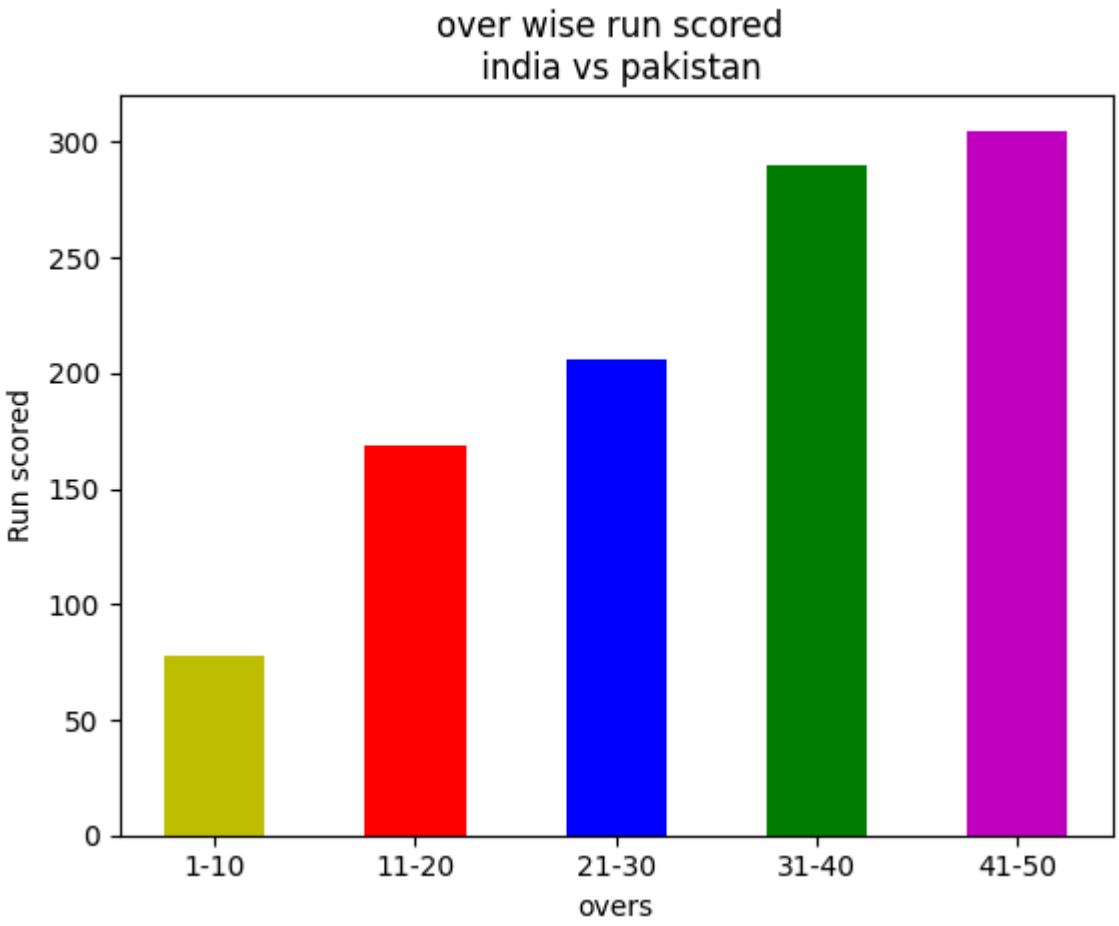
```
   empid     ename         doj
0    101     khush  12-01-2022
1    102    khushi   1-01-2022
2    103      jiya   2-08-2012
3    104  zankhana  12-08-2002
4    105     vidhi   2-05-2003
5    106   prinshi  12-01-2022
   empid   ename         doj
0    101   khush  12-01-2022
1    102  khushi   1-01-2022
   empid    ename         doj
4    105    vidhi   2-05-2003
5    106  prinshi  12-01-2022
   empid     ename         doj
2    103      jiya   2-08-2012
3    104  zankhana  12-08-2002
4    105     vidhi   2-05-2003
```

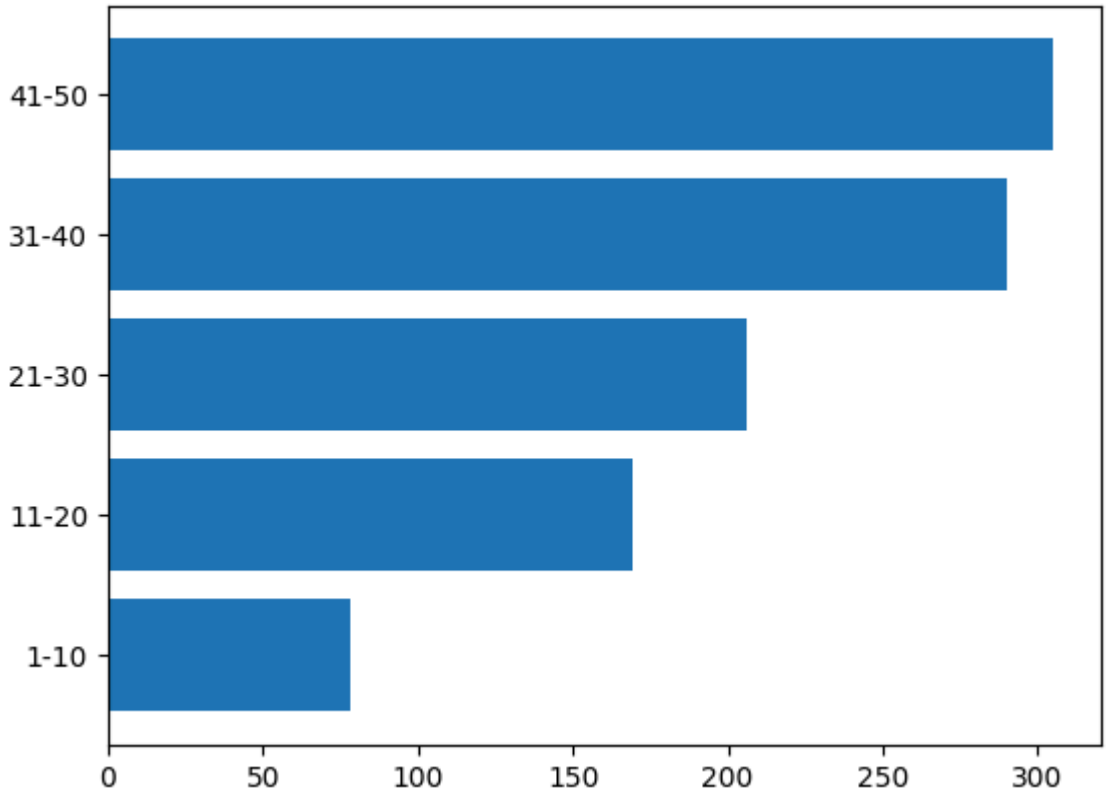
In [11]:

```
import matplotlib.pyplot as plt
x=[1,2,3]
y=[2,3.5,5]
plt.plot(x,y)
plt.show()
```



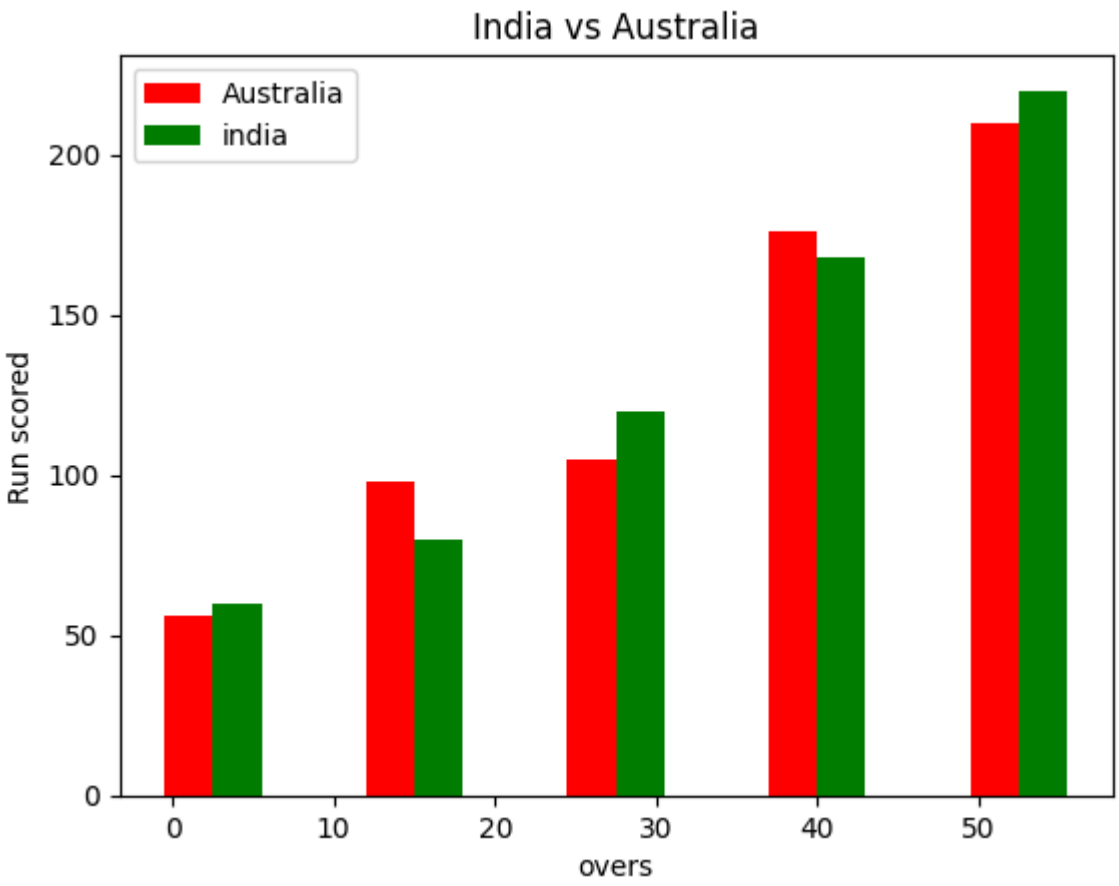
In [ ]:
```
#Bar graph
from matplotlib.pyplot import*
from numpy import*
x=['1-10','11-20','21-30','31-40','41-50']
y=[78,169,206,290,305]
xlabel('overs')
ylabel('Run scored')
title('over wise run scored \n india vs pakistan')
bar(x,y,color=['y','r','b','g','m'],width=0.5)    #defalt width=0.5
show()

print('\n')
barh(x,y)
show()
```



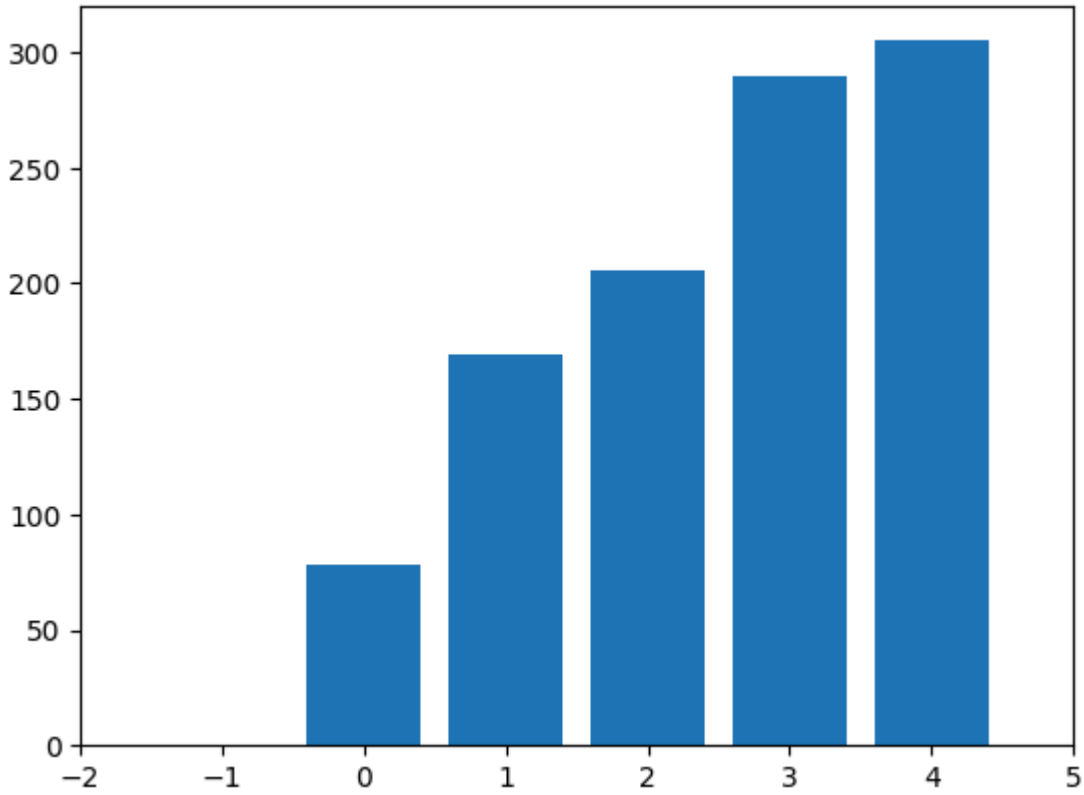


In [ ]:
```
#Multiple bar
from matplotlib.pyplot import*
from numpy import*
a=[56,98,105,176,210]
b=[60,80,120,168,220]
```

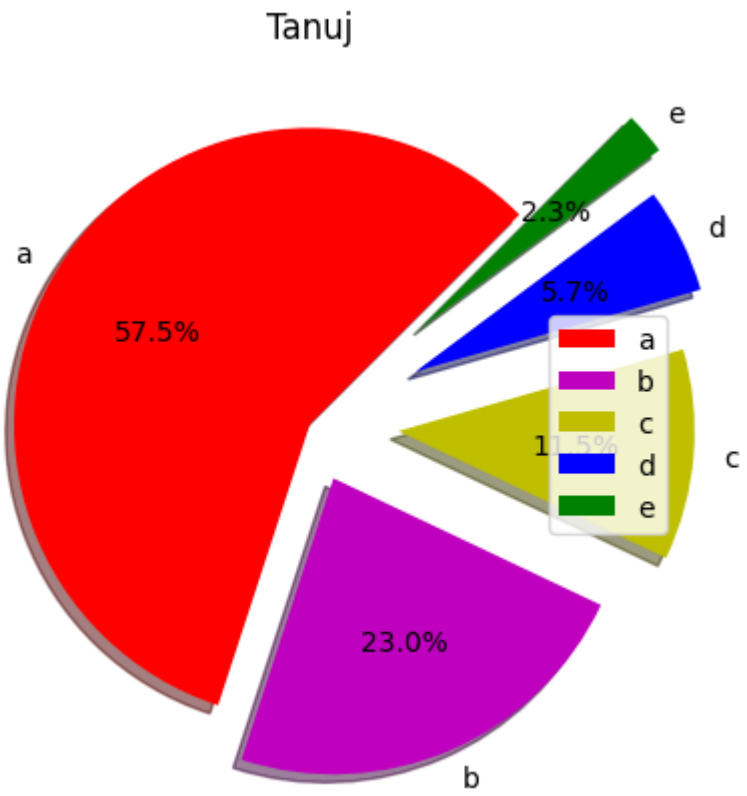
```
x=linspace(1,51,5)
bar(x,a,width=3,color='r',label='Australia')
bar(x+3,b,width=3,color='g',label='india')
xlabel('overs')
ylabel('Run scored')
title('India vs Australia')
legend()
show()
```



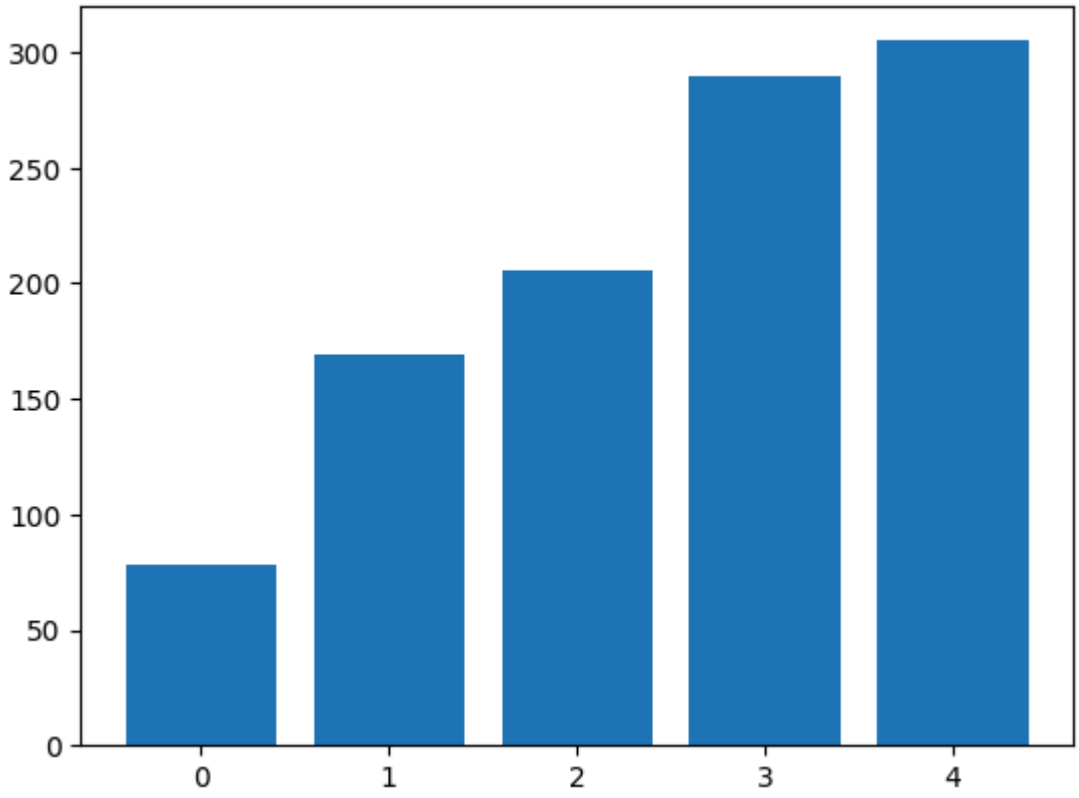
In [ ]:
```
#limit in bar chart
from matplotlib.pyplot import*
from numpy import*
x=arange(5)
xlim(-2.0,5.0)
bar(x,y)
show()
```



In [ ]:
```
#Pie chart
from matplotlib.pyplot import*
s=[50,20,10,5,2]
d=['a','b','c','d','e']
c=['r','m','y','b','g']
e=[0,0.2,0.3,0.4,0.5]
pie(s,labels=d,colors=c,startangle=45,explode=e,shadow=True,autopct='%.1f
title('khush')
legend()
show()
```



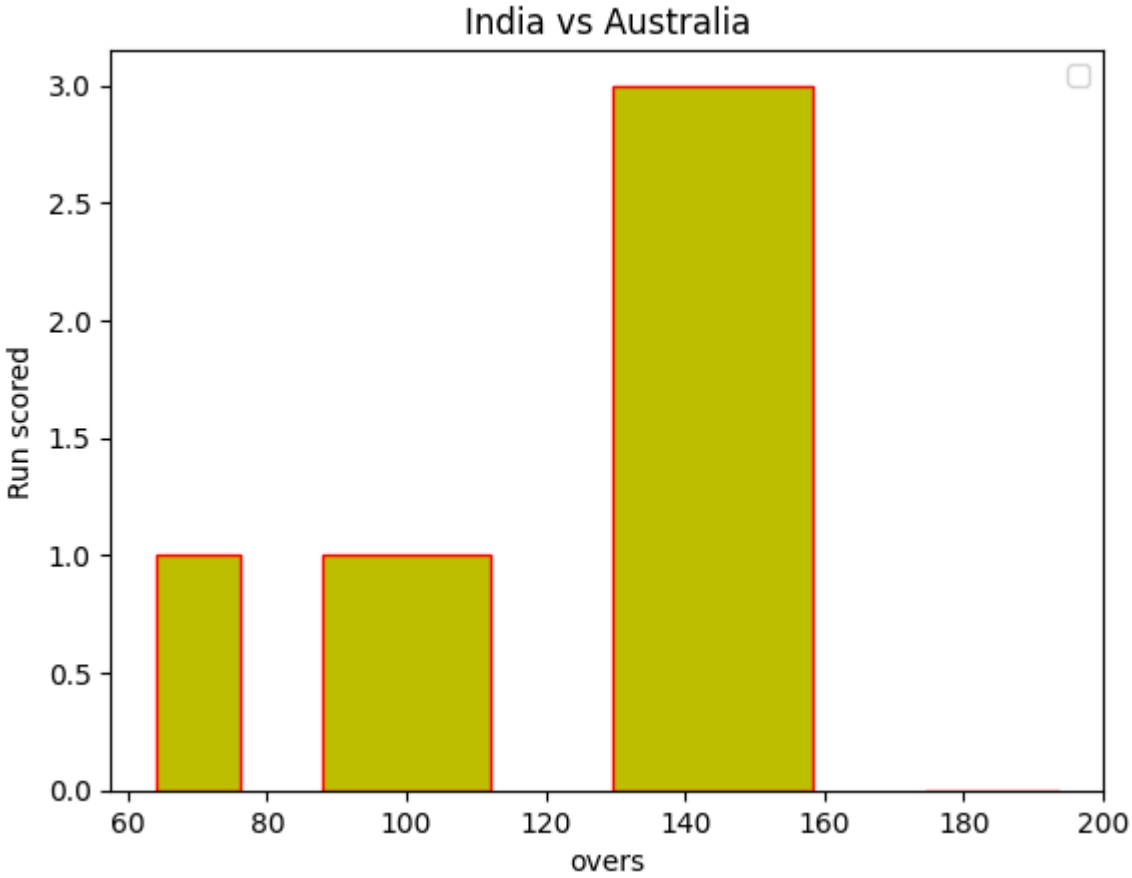
In [ ]:
```
#stick
from matplotlib.pyplot import*
from numpy import*
x=arange(5)
xticks([0,1,2,3,4])                #by defalt=0.5
bar(x,y)
show()
```



In [ ]:
```
#Histrogram
from matplotlib.pyplot import*
```

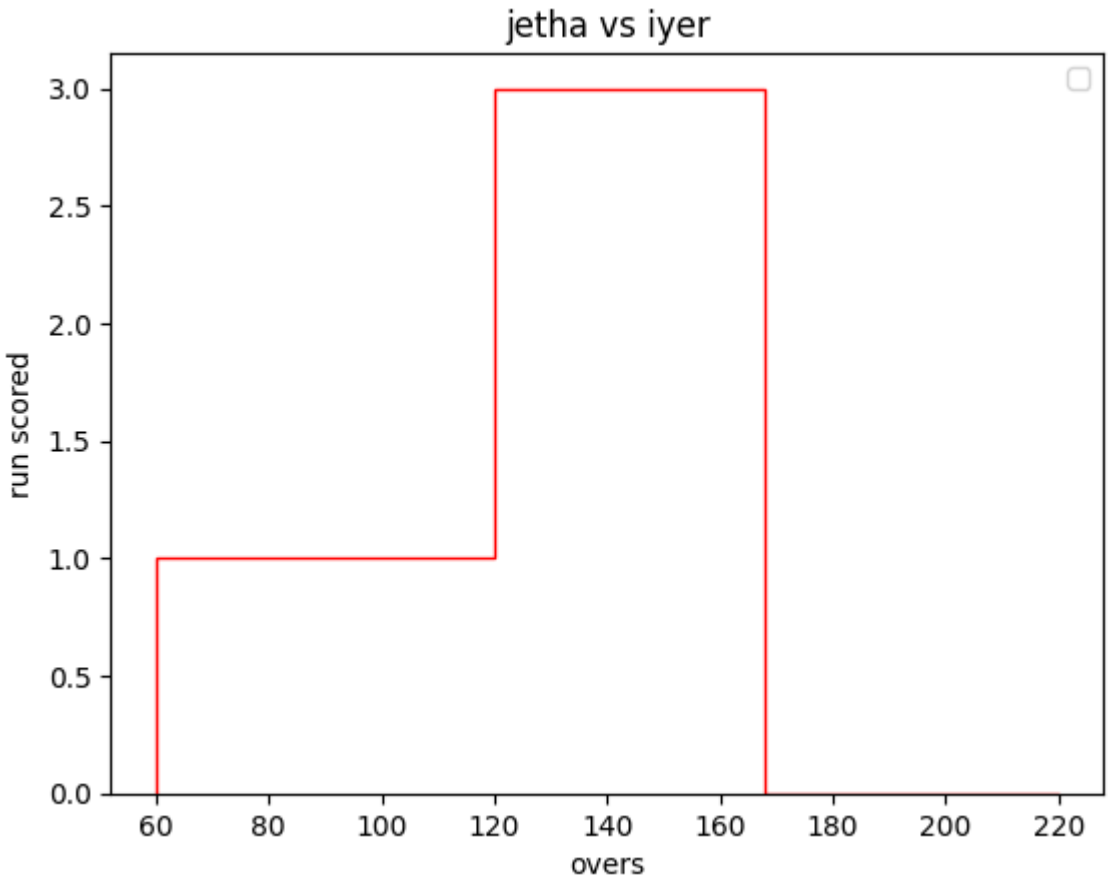
```
a=[70,90,130,140,150]
b=[60,80,120,168,200]
hist(a,bins=b,color='y',rwidth=0.6,edgecolor='r')
xlabel('overs')
ylabel('Run scored')
title('India vs Australia')
legend()
show()
```

```
<ipython-input-14-3e2a38eb24a4>:9: UserWarning: No artists with labels fou
nd to put in legend.  Note that artists whose label start with an undersco
re are ignored when legend() is called with no argument.
  legend()
```



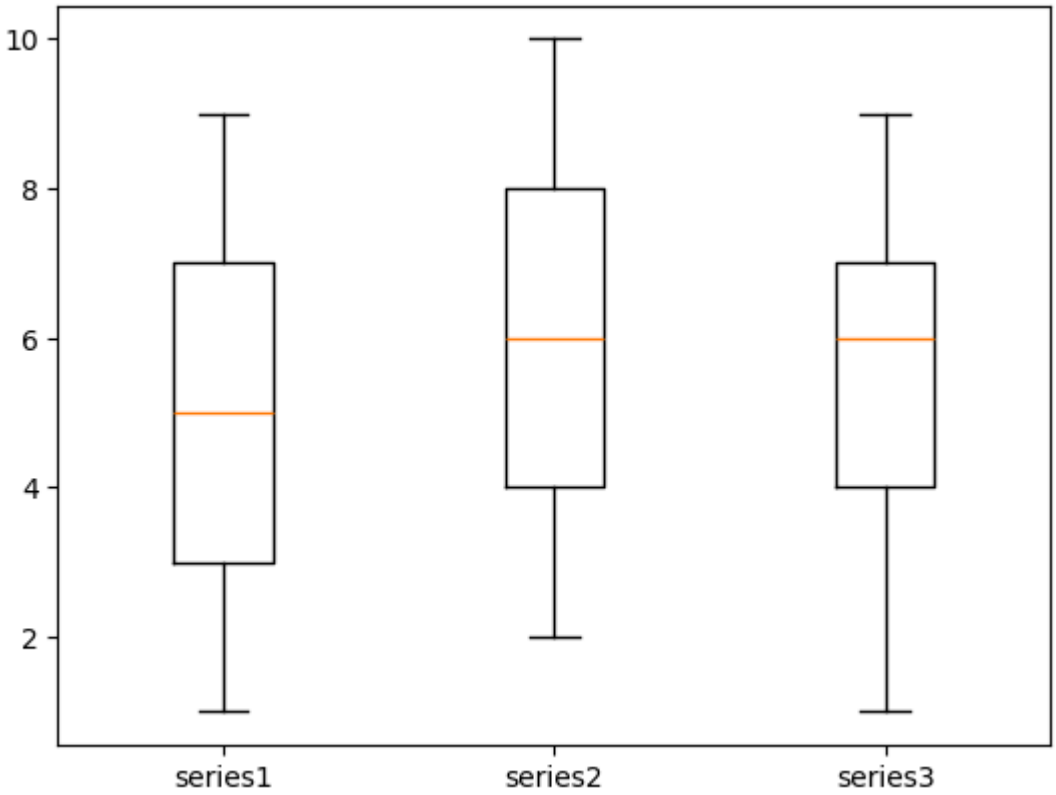
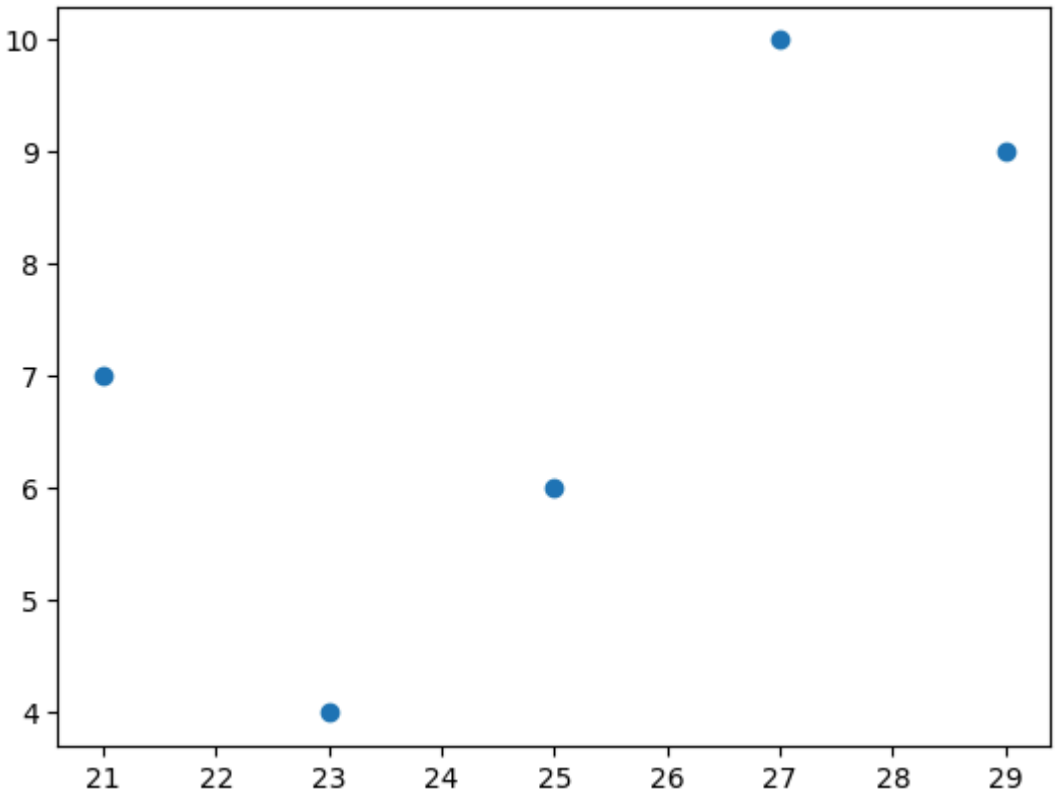
In [ ]:
```
#frequency histogram
from matplotlib.pyplot import *
a=[70,90,130,140,150]
b=[60,80,120,168,220]
hist(a,bins=b,color='y',histtype='step',rwidth=0.6,edgecolor='r')
xlabel('overs')
ylabel('run scored')
title('jetha vs iyer')
legend()
show()
```

```
<ipython-input-20-39c55e2cc20a>:9: UserWarning: No artists with labels fou
nd to put in legend.  Note that artists whose label start with an undersco
re are ignored when legend() is called with no argument.
  legend()
```



In [ ]:
```
#Box plot
from matplotlib.pyplot import*
v1=[1,3,5,7,9]
v2=[2,4,6,8,10]
v3=[1,4,6,7,9]
data=[v1,v2,v3]
boxplot(data,labels=['series1','series2','series3'])
show()
```

```
<ipython-input-21-4c67302e309a>:7: MatplotlibDeprecationWarning: The 'labe
ls' parameter of boxplot() has been renamed 'tick_labels' since Matplotlib
3.9; support for the old name will be dropped in 3.11.
  boxplot(data,labels=['series1','series2','series3'])
```



In [ ]:
```
#Scatter graph
from matplotlib.pyplot import*
x=[23,25,21,27,29]
y=[4,6,7,10,9]
scatter(x,y)
show()
```